Data Analytics with Tableau

ASSIGNMENT 1

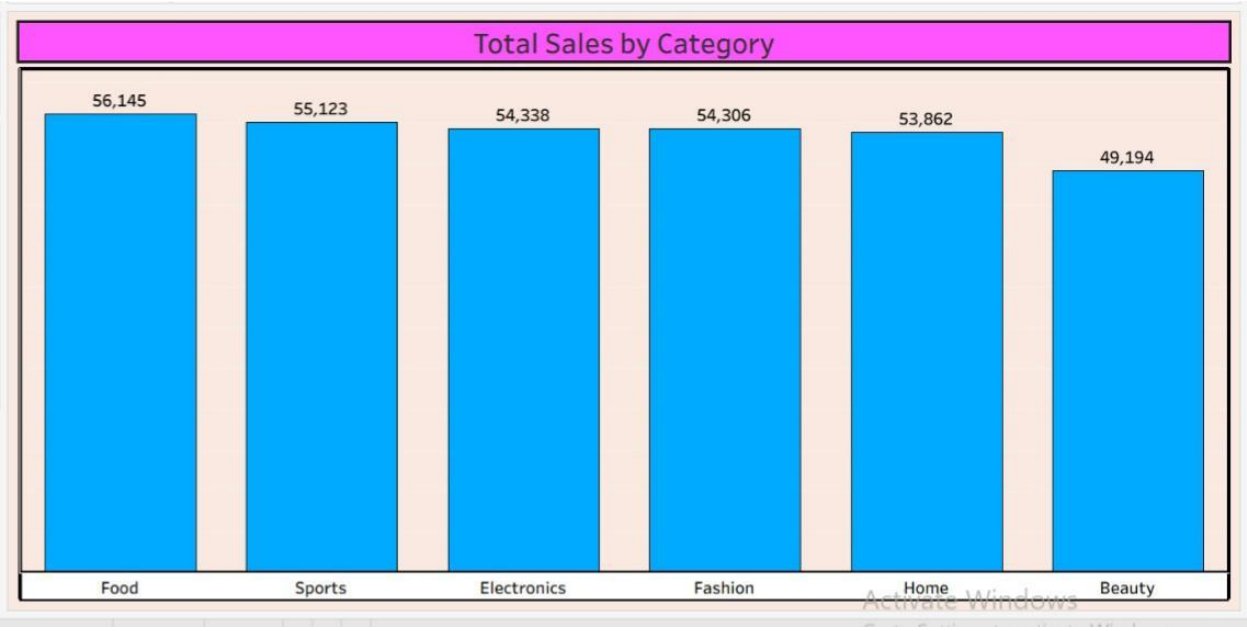
Aim: The growth of supermarkets in the most populated cities is increasing and market Competitions are also high. The dataset is one of the historical sales of supermarket Company which has recorded in 3 different branches for 3 months data.
Creating below visualization

BAR CHART: TOTAL SALES BY CATEGORY

Purpose: A bar chart is used to represent data visually using rectangular bars. It helps values across different categories easily.

STEPS: Open Tableau Public, Load dataset. Click Sheet 1.

- Next, Drag Product line → to Columns and Drag Total (Sales/Amount) → to
- Rows (vertical bars will automatically appear)
- In the Marks card , select Bar.
- On the Marks card → Click Label. Tick Show Mark Labels. Now numbers (56,145 etc.) appear on top of bars.
- Sort Bars (Descending) using sort icon.
- Formatting : Change Bar Color Click Color (Marks card) Choose Blue .
- Change Background Color, Go to Format → Shading.
- Under Worksheet → Pane → Select light green (like screenshot).
- Format Title, Double-click title give title as Branch performance segmentation Click OK.
- To change Color-Go to Format → Font → Title → Choose Color (Green/ sandel).
- Go to Format →Give Borders

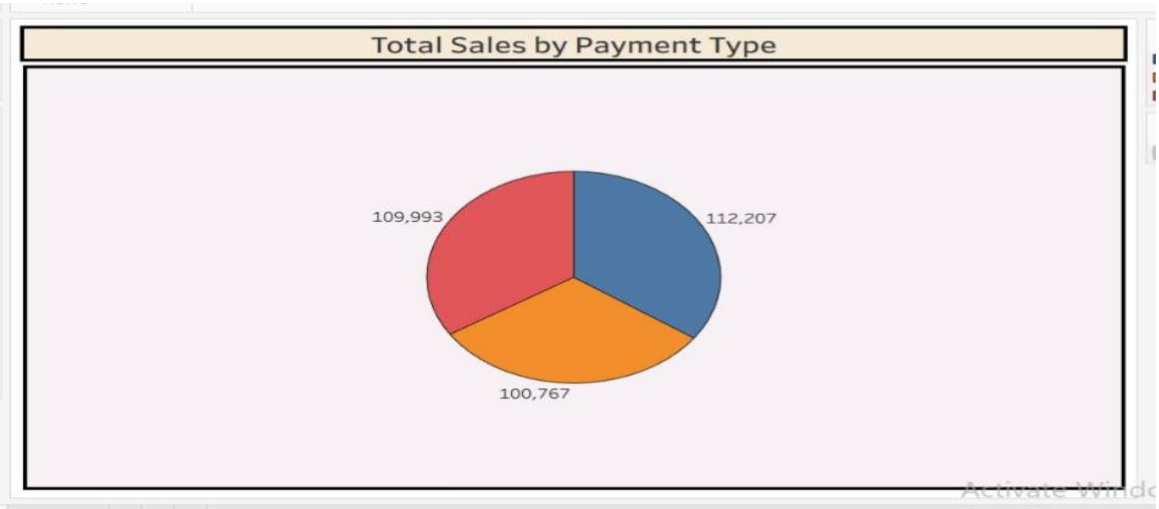


2.PIC CHAT: TOTAL SALES BY PAYMENT TYPE

Purpose: A pie chart is used to represent data as parts of a whole. It shows how each category contributes to the total using slices of a circle.

STEPS: Open Tableau Public, Load dataset. Click Sheet 2.

- Go to Marks Card, Select Pie.
- Drag Payment to the colour in the mark Card.
- Drag Payment Again to Label, right click label and Tick Show mark Label.
- Drag CNT(payment) To angle and Label.
- For percentage, go Show Percentage on Total on label.
- Change Background Color, Go to Format → Shading.
- Under Worksheet → Pane → Select light green (like screenshot).
- Format Title, Double-click title give title as market share breakdown Click OK.
- To change Color-Go to Format → Font → Title → Choose Color(pink / light green)
- Go to Format →Give Borders

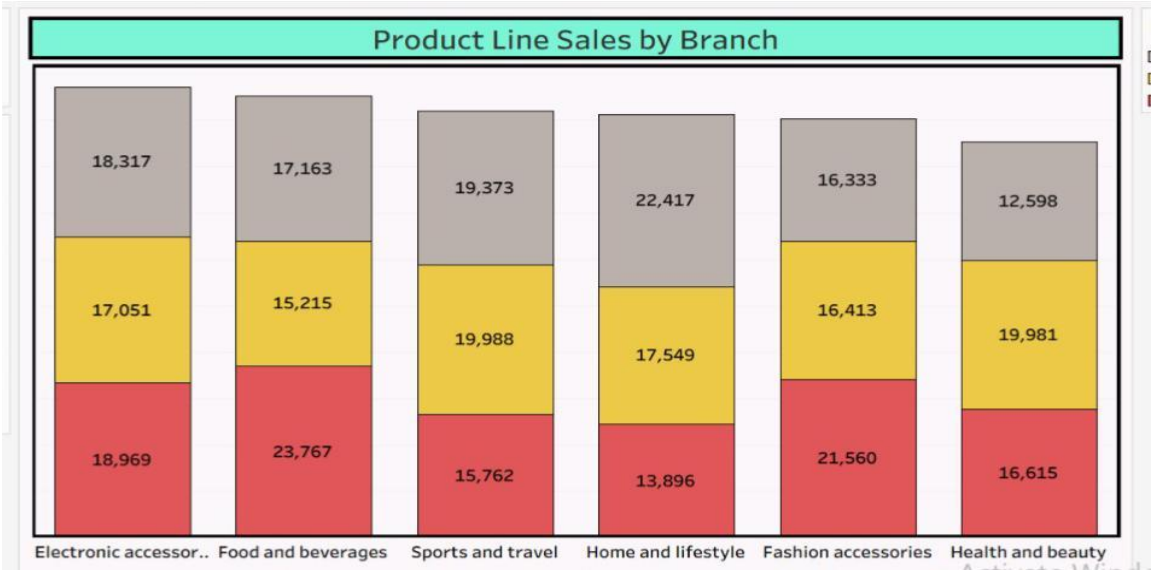


3.STACKED BAR CHART:PRODUCT LINE SALES BY BRANCH

Purpose: A stacked bar chart is used in branch-wise sales to compare total sales of each branch and also see the contribution of different products or categories within that branch.

STEPS: Open Tableau Public, Load dataset. Click Sheet 3.

- Drag Branch to Columns And Drag Total (or Sales/Amount) to Rows .
- Now see simple vertical bars for each branch.
- Go to Marks card (left side), Click dropdown (Automatic) Select Bar.
- Drag Gender to Color (in Marks card).(Now bars split into Male & Female colors.)
- Drag Gender to Label (Marks card). Then Right Click Label Show mark labels.
- Go to Format → Shading. Under Worksheet → Pane → Select Colour
- Format Title, Double-click title give title as proportional contribution bars and Click OK.
- Go to Format → Font → Title → Choose Colour ,Then Go to border and give border.

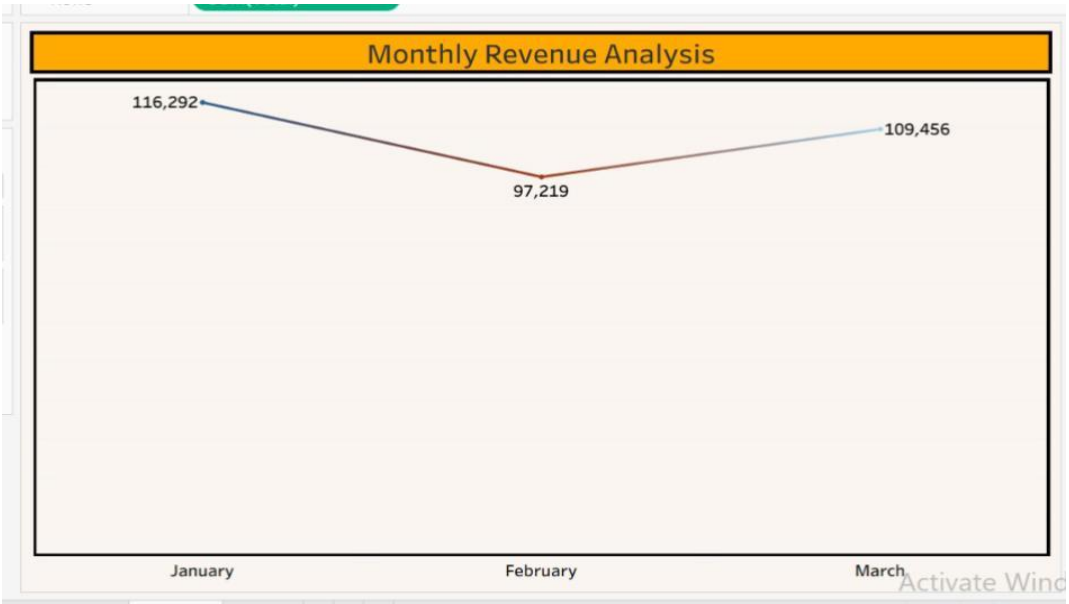


4.LINE CHART: MONTHLY REVENUE ANALYSIS

Purpose: A line chart for monthly data is used to show how values change from month . to month across a year or any time period.

STEPS: Open Tableau Public, Load dataset. Click Sheet 4.

- Drag Date to Columns and Select MONTH(Date).
- Drag Sum Total to Rows.
- Go to Marks Select Line Then Drag SUM(Total) to Label and Tick Show Mark Labels, Automatically Values appear on line.
- Format Line Color Choose single Color and Size Increase thickness.
- Go to Format → Shading. Under Worksheet → Pane → Select Colour
- Format Title, Double-click title give title as performance Trend line and Click OK.
- Go to Format → Font → Title → Choose Colour ,Then Go to borders and give borders

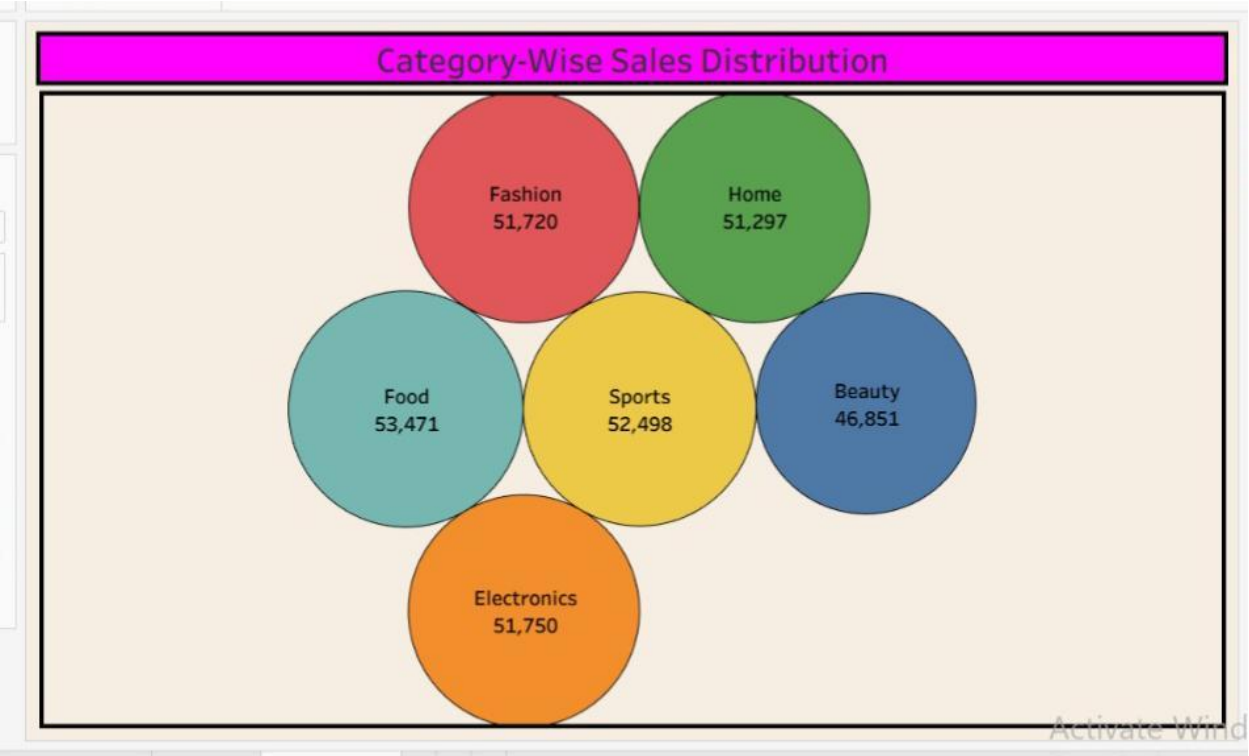


5.BUBBLE CHART: CATEGORY WISE SALES DISTRIBUTION

Purpose: A Bubble Chart is used to visualize and compare product line performance using circles (bubbles).Each bubble represents one product category.

STEPS: Open Tableau Public, Load dataset. Click Sheet 5.

- Go to the Marks card and change from Automatic Circle.
- This converts the view into a bubble-style chart or select bubble chart from Show me.
- Drag Product line Label And Drag SUM(Quantity) to Size.
- Increase the Size slider to make bubbles more visible.
- Click Label and Tick Show Mark Labels.
- Format Axes for Right-click Unit price axis → Edit Axis and Give Title Unit Price
- Drag SUM(unit price) to colour
- Go to Format → Shading. Under Worksheet → Pane → Select Colour .
- Format Title, Double-click title give title as revenue constellation and Click OK.
- Go to Format → Font → Title → Choose Colour ,Then Go to borders and give borders



Assignment 2

DONUT CHART: Order distribution by shipping type

Purpose: To represent part-to-whole relationships by showing how each category contributes to the total value.

Steps: Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset → Go to Sheet 1.

- Drag Category to Color (Marks card) then Drag Total to Angle and Drag Total → Label
- Change Marks type → Pie and For Create Dual Pie , Drag Number of Records → Rows (twice) Right-click second axis → Dual Axis Click 2nd Marks card (AVG(Number of Records)) and Remove Category from Color
- Change Color to White (or background Color) and Reduce Size Make it smaller
- Go to 2nd Marks card, Drag Number of Records to Label
- Edit label to Increase font size and Align Center
- Go to Format → Shading. Under Worksheet → Pane → Select Colour.
- Format Title, Double-click title give title as customer segment distribution and Click OK.
- Go to Format → Font → Title → Choose Colour , Then Go to borders and give borders

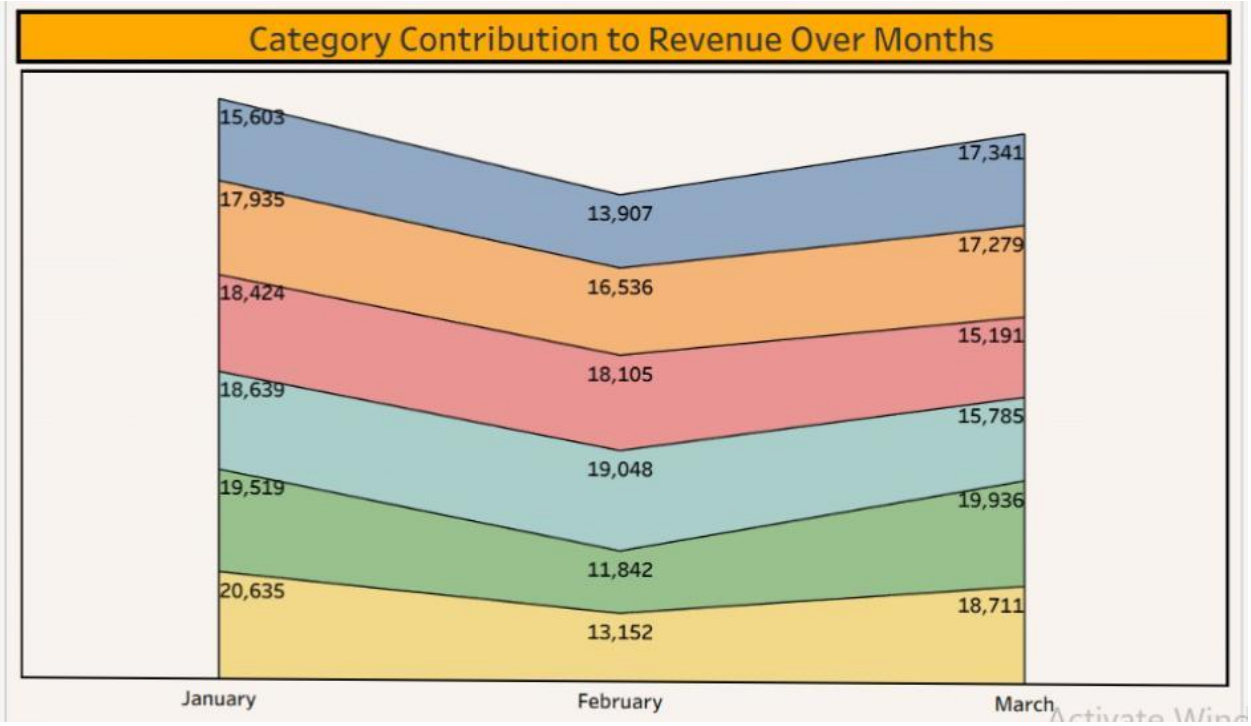
AREA CHART: CATEGORY CONTRIBUTION TO REVENUE OVER MONTHS

Purpose: To show how different categories contribute to the total over time and compare their trends and growth.

Steps:Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset → Go to Sheet 2.

- Drag Date → Columns → Change to MONTH(Date)
- Drag Total → Rows
- Drag City → Rows (place before SUM(Total))
- Change Marks type → Area
- Drag Branch → Color (Marks card)
- Adjust Size if needed from Size option
- Go to Format → Shading. Under Worksheet → Pane → Select Colour.

- Format Title, Double-click title give title as growth coverage map and Click OK.
- Go to Format → Font → Title → Choose Colour , Then Go to borders and give borders



3. TEXT TABLE: CATEGORY -WISE SALES TEXT TABLE

Purpose: To display exact data values in a tabular format for detailed comparison and reporting.

Steps: Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset → Go to Sheet 3.

- From Data pane Drag Branch to Rows Then Drag Total → to Text (Marks card)
- Go to Marks dropdown Select Text
- Right-click SUM(Total) Click Format , Under Numbers Choose Currency (Custom) , Display Units Thousands (K)
- Go to Format → Shading. Under Worksheet → Pane → Select Colour.
- Format Title, Double-click title give title as data insight
- Go to Marks card → Change Marks type → Gantt Bar.
- Drag SUM(Gross Income) to Size
- Drag SUM(Gross Income) to Color.
- Go to Format → Shading. Under Worksheet → Pane → Select Colour.
- Format Title, Double-click title give title as profit flow dynamics and Click OK.
- Go to Format → Font → Title → Choose Colour Then to borders and give borders.

CATEGORY-WISE SALES TEXT TABLE					
City	Product line	January	February	March	Grand Total
Mandalay	Electronic acces..	6.4K	6.4K	3.5K	16.2K
	Fashion accesso..	5.8K	5.8K	4.0K	15.6K
	Food and bevera..	6.3K	5.3K	2.9K	14.5K
	Health and beau..	6.1K	5.6K	7.4K	19.0K
	Home and lifesty..	4.4K	4.4K	7.9K	16.7K
	Sports and travel	6.4K	5.3K	7.3K	19.0K
Naypyitaw	Electronic acces..	5.5K	5.2K	7.4K	18.1K
	Fashion accesso..	6.1K	7.3K	7.1K	20.5K
	Food and bevera..	7.9K	7.0K	7.7K	22.6K
	Health and beau..	5.7K	5.6K	4.5K	15.8K
	Home and lifesty..	5.3K	2.9K	5.0K	13.2K
	Sports and travel	8.0K	3.4K	3.7K	15.0K
Yangon	Electronic acces..	6.1K	5.0K	6.4K	17.4K
	Fashion accesso..	6.5K	4.9K	4.1K	15.6K
	Food and bevera..	4.4K	6.7K	5.2K	16.3K
	Health and beau..	3.8K	2.8K	5.4K	12.0K
	Home and lifesty..	9.8K	4.5K	7.0K	21.3K
	Sports and travel	6.2K	4.5K	7.7K	18.5K

4. HIGHLIGHTED TABLE: Monthly revenue heatmap table

Purpose:To compare sales performance across product segments using color intensity to identify high and low values quickly.

Steps:Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset → Go to Sheet 4.

- Drag Product line to Columns Then Drag Branch to Rows.
- Drag Total to Color (Marks card) Next Drag Total Label
- Change Marks type Square (Or click Show Me → Highlight Table)
- Click Label → Show mark labels Format numbers to Currency
- Go to Format → Shading. Under Worksheet → Pane → Select Colour.
- Format Title, Double-click title give title as kpi highlight dashboard and Click OK.
- Go to Format → Font → Title → Choose Colour , Then Go to borders and give borders

Monthly Revenue Heatmap Table				
Categ..	January	February	March	Grand Total
Food	932	952	789	2,674
Sports	1,032	658	936	2,625
Electronics	897	827	864	2,588
Fashion	921	905	760	2,586
Home	976	592	997	2,565
Beauty	780	695	867	2,343

5. WORD CLOUD: PRODUCT LINE -WISE SALES WORD CLOUD

Purpose:To visualize text data by displaying frequently occurring words in larger and more prominent sizes.

Steps:

- Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset → Go to Sheet 5.
- Go to Marks card, Select Text.
- Drag Product line to Text (Marks card) Now product names appear as words.
- Drag Quantity (or Sales / Total) to Size.
- Drag Product line to Color (Now each word has a different color).
- Click Size , Move slider → Increase for big bold words.
- Go to Format → Shading. Under Worksheet → Pane → Select Colour.
- Format Title, Double-click title give title as data expression cloud and Click OK.
- Go to Format → Font → Title → Choose Colour
- Then Go to borders and give borders

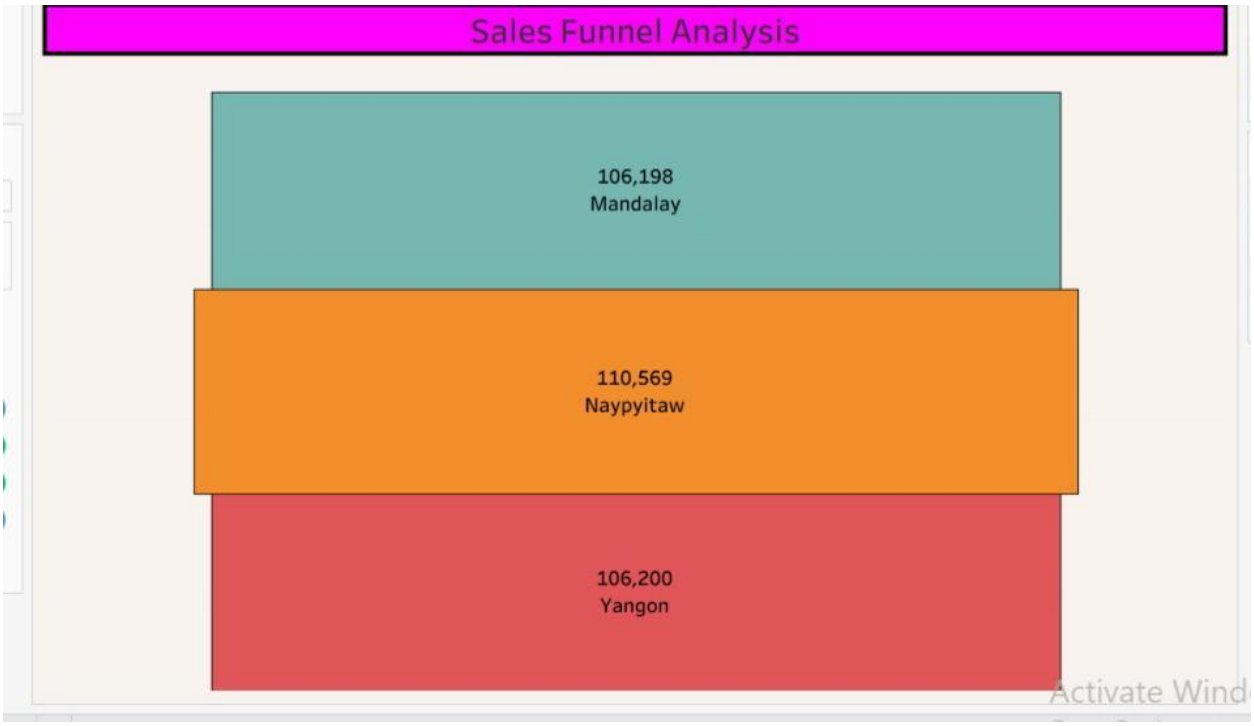
6. FUNNEL CHART: SALES FUNNEL ANALYSIS

Purpose: A Funnel Chart is used to show how data moves through different stages and how the value decreases step-by-step.

Steps:Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset → Go to Sheet 6.

- Drag Product line to Rows. Then Drag SUM(Total) Columns. Now Horizontal bars appear.

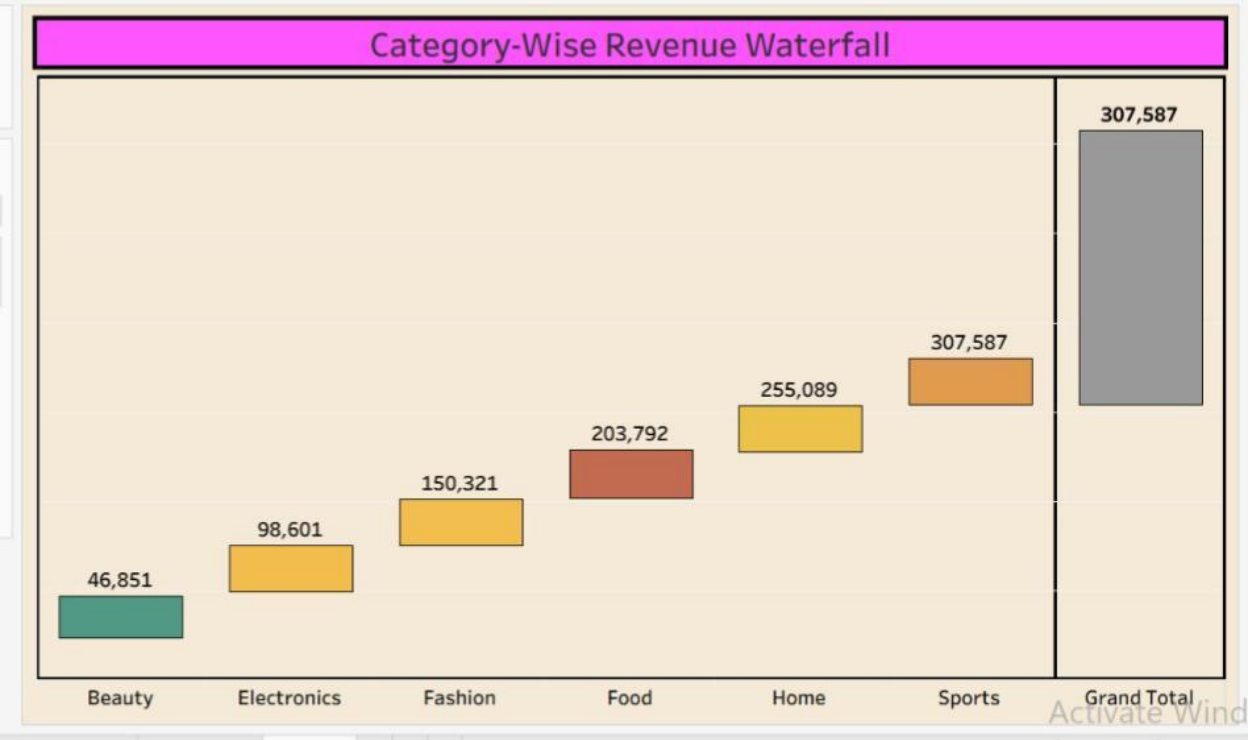
- Click Sort Descending icon
- Now the biggest bar on top (funnel shape order).
- Go to the Marks card, Select Bar.
- Drag Product line to Color. Now each bar gets a different color.
- Click Size, Drag slider for Increase width.
- Go to Format → Shading. Under Worksheet → Pane → Select Colour.
- Format Title, Double-click title give title as demand conversation funnel and Click OK.
- Go to Format → Font → Title → Choose Colour
- Then Go to borders and give borders



7. WATERFALLS: CATEGORY -WISE REVENUE WATERFALL

Purpose: A Waterfall Chart shows how an initial value changes step-by-step due to increases and decreases. It explains how the final value is reached.

Steps:Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset → Go to Sheet 7.

- Drag Date to Columns and Change to MONTH(Date).
- Drag Gross Income to Rows.
- Right-click SUM(Gross Income) , Select Quick Table Calculation → Running Total.
- Go to Marks card → Change Marks type → Gantt Bar.
- Drag SUM(Gross Income) to Size
- Drag SUM(Gross Income) to Color.
- Go to Format → Shading. Under Worksheet → Pane → Select Colour.
- Format Title, Double-click title give title as profit flow dynamics and Click OK.

- Go to Format → Font → Title → Choose Colour.
- Then go to borders and give borders.